CS 306: Network Models

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Outline

- Protocol Layering
- OSI Model
- TCP/IP Protocol

Protocol Layering

- A protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.
- When communication is simple, we may need only one simple protocol; when the communication is complex, we need a protocol at each layer, or protocol layering.

Principles of Protocol Layering

- The first principle dictates that if we want bidirectional communication, we need to make each layer so that it can perform two opposite tasks, one in each direction.
- The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.

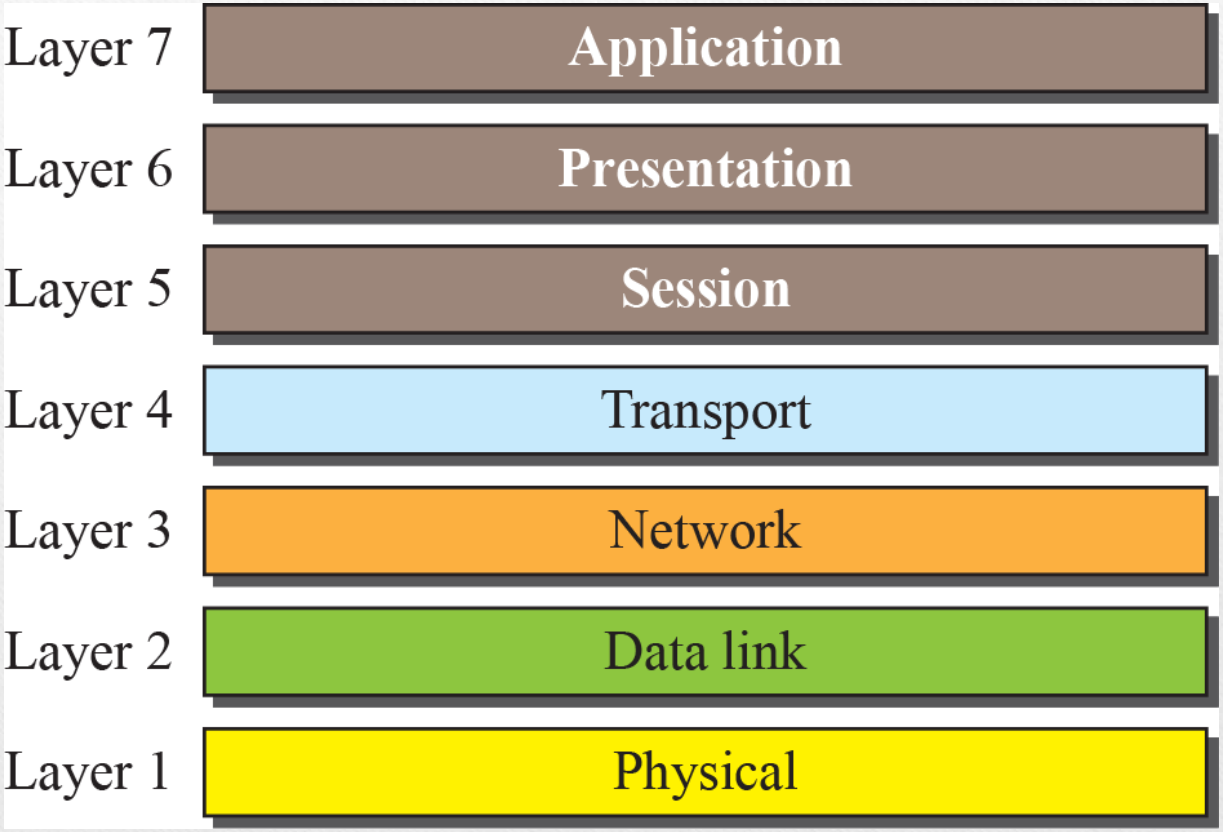
Logical Connections

- We can think about logical connection between each layer. This means that we have layer-to-layer communication.
- This logical (imaginary) connection at each layer enables for sending the object created from that layer.
- We will see that the concept of logical connection will help us better understand the task of layering we encounter in data communication and networking.

OSI Model

- Open Systems Interconnection model (OSI model) is a conceptual model that characterizes the standard communication functions of computing system
- It was developed by the International Organization for Standardization (ISO)

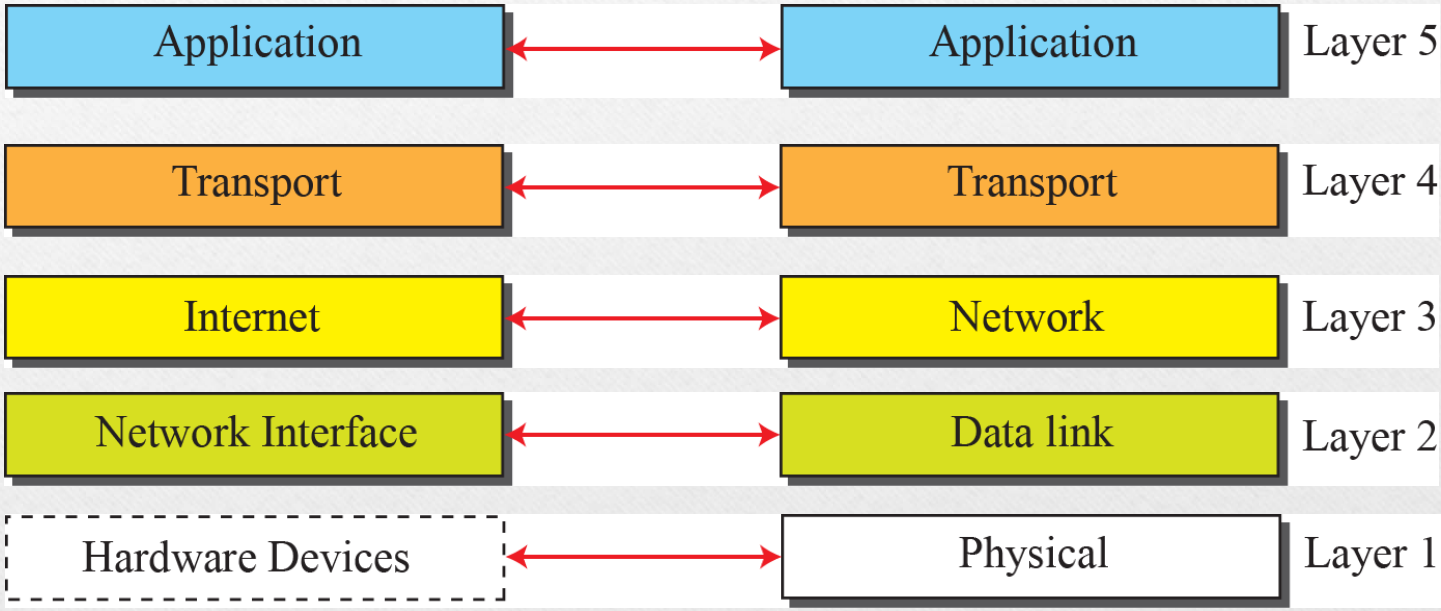
Layers in OSI Model



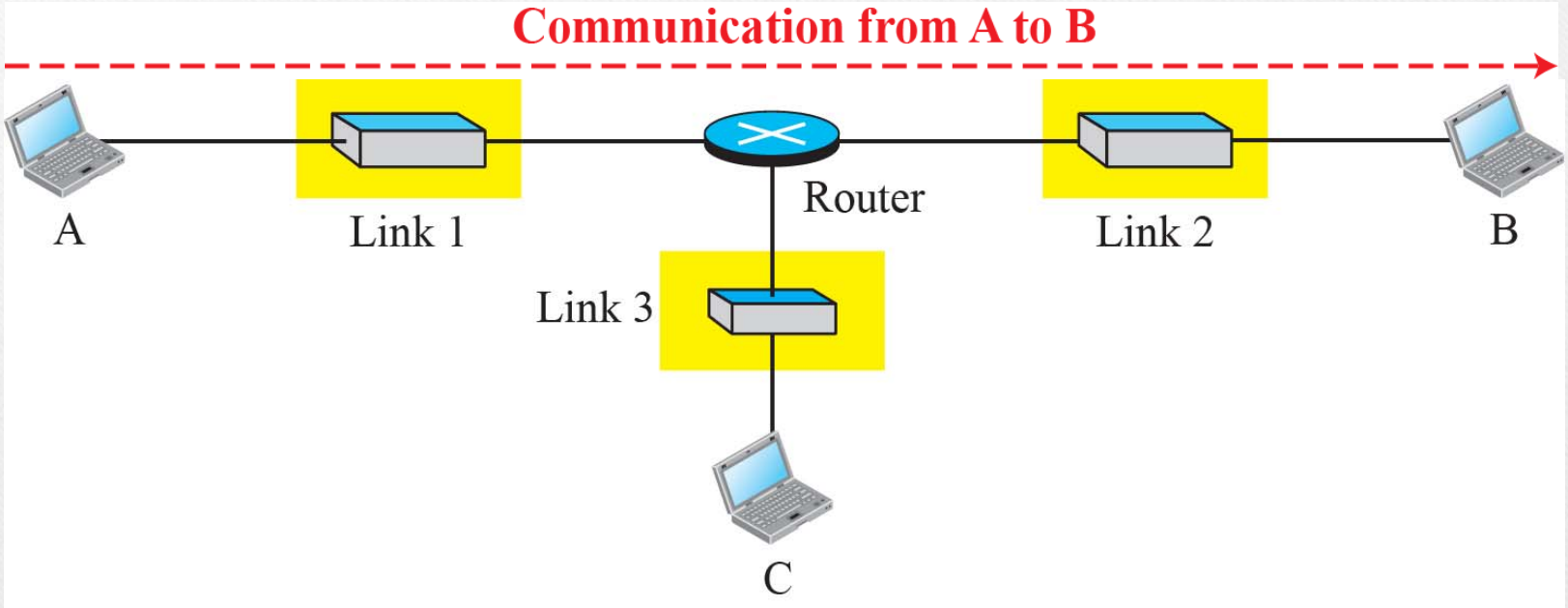
TCP/IP Protocol

- Transmission Control Protocol (TCP) and Internet Protocol (IP) are two distinct computer network protocols.
- TCP and IP are so commonly used together, that TCP/IP has become standard terminology
- IP is the primary way in which network connections are made, and it establishes the basis of the Internet.
- TCP handle packet ordering or error checking functionality.

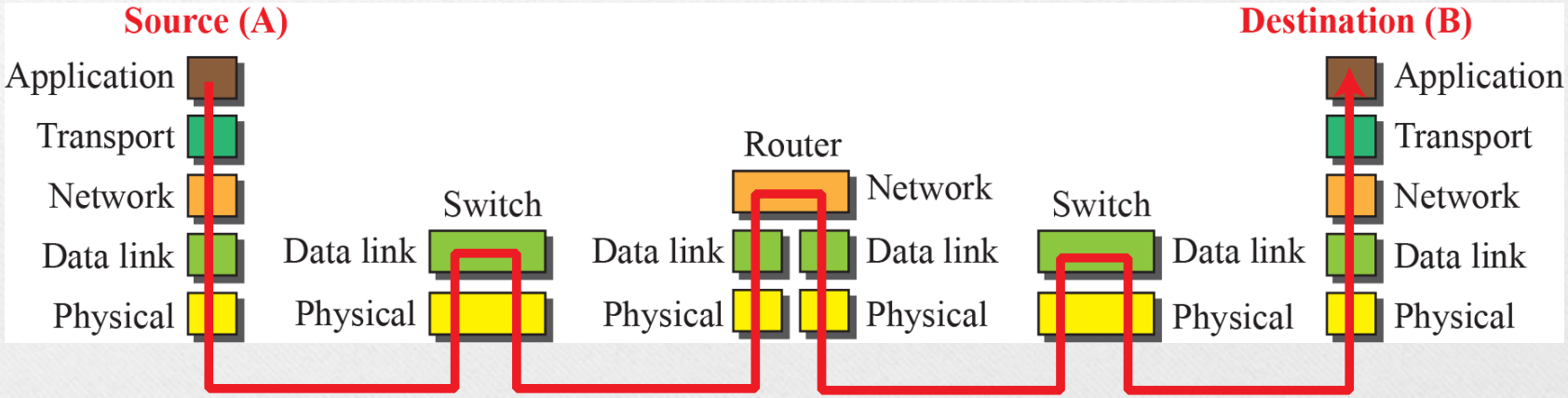
Layers in TCP/IP Protocol



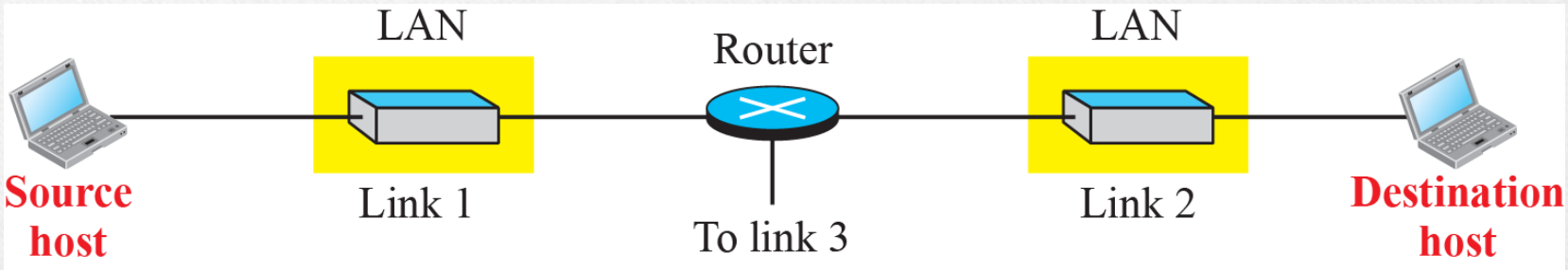
Layered Architecture (Example)



Layered Architecture (Example) (Contd.)

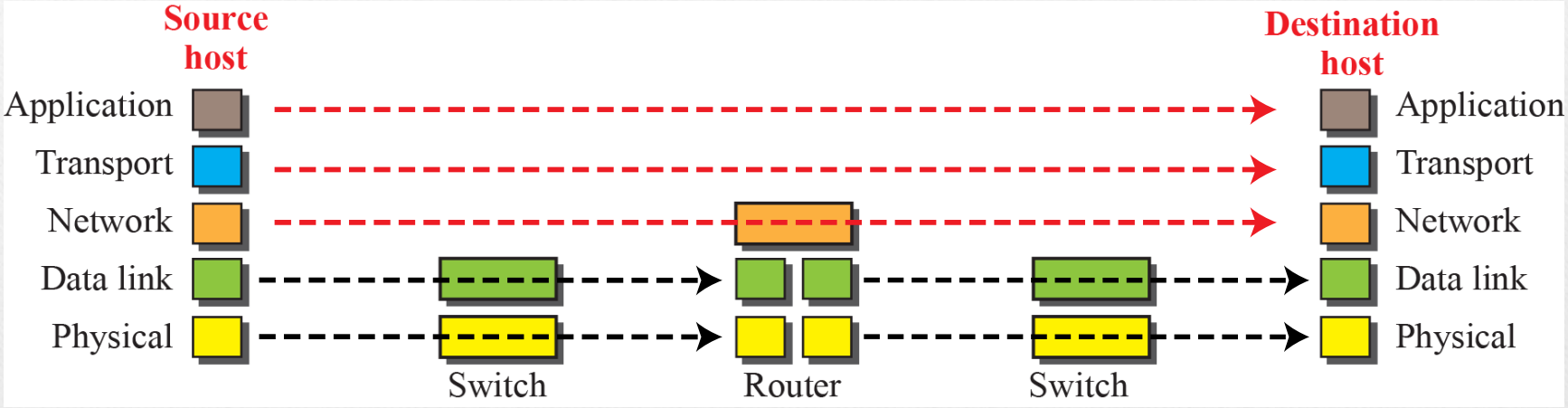


Logical connections in TCP/IP

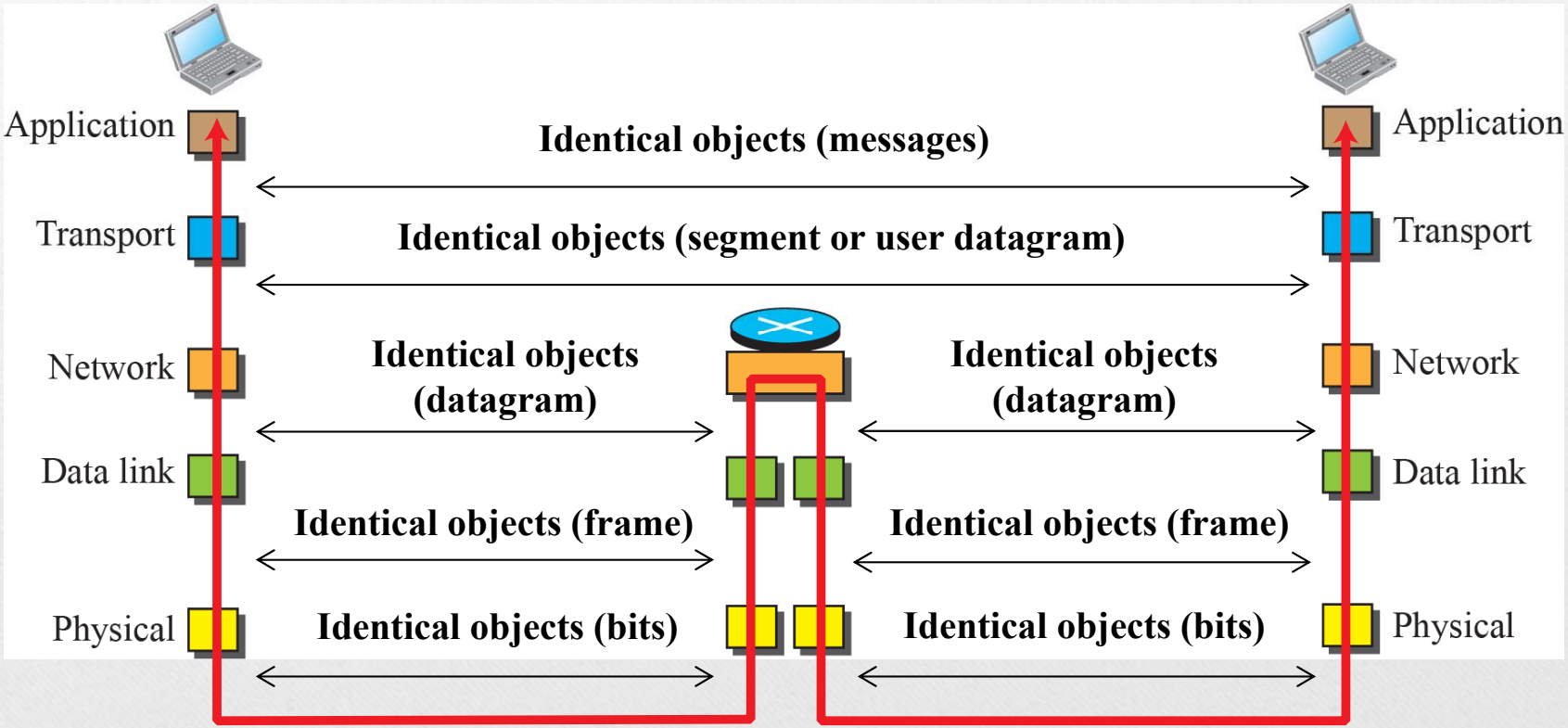


Logical connections in TCP/IP (Contd.)

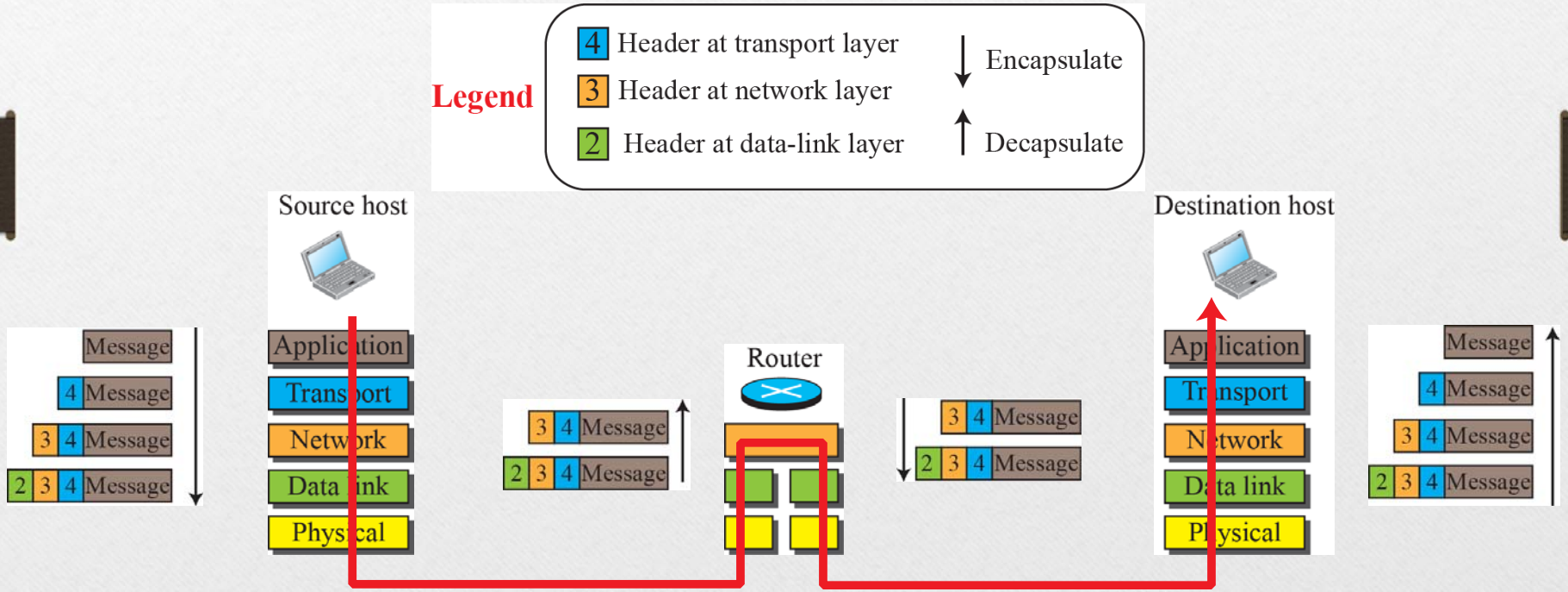
Logical connections



Logical connections in TCP/IP (Contd.)



Encapsulation / Decapsulation



Addressing in TCP/IP

- Any communication that involves two parties needs two addresses: source address and destination address.
- The physical layer does not need addresses
- Therefore, we have four type of addressing, i.e., one pair per layer.

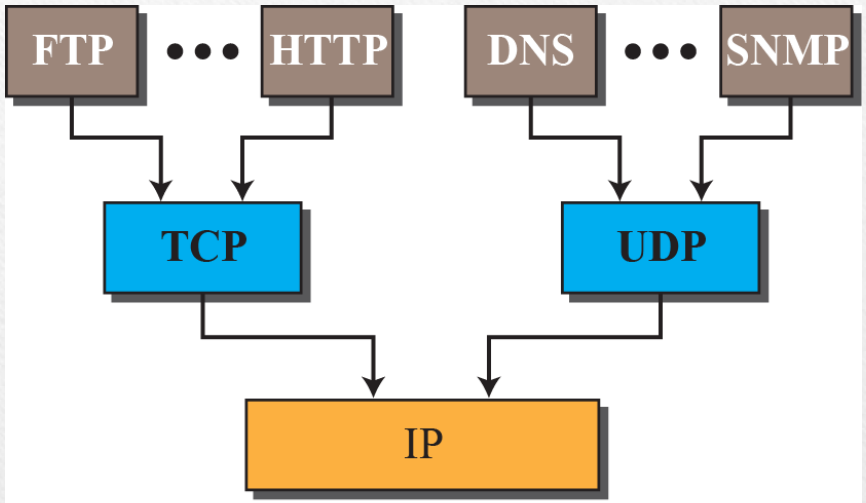
Addressing in TCP/IP (Contd.)

Packet names	Layers	Addresses
Message	Application layer	Names
Segment / User datagram	Transport layer	Port numbers
Datagram	Network layer	Logical addresses
Frame	Data-link layer	Link-layer addresses
Bits	Physical layer	

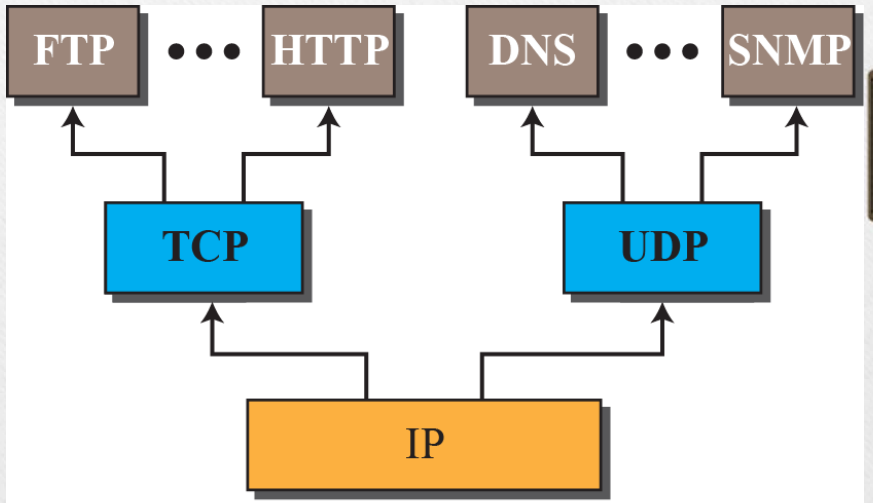
Multiplexing and Demultiplexing

- Multiplexing in this case means that a protocol at a layer can encapsulate a packet from several next-higher layer protocols (one at a time)
- Demultiplexing means that a protocol can decapsulate and deliver a packet to several next-higher layer protocols (one at a time).

Multiplexing and Demultiplexing (Contd.)

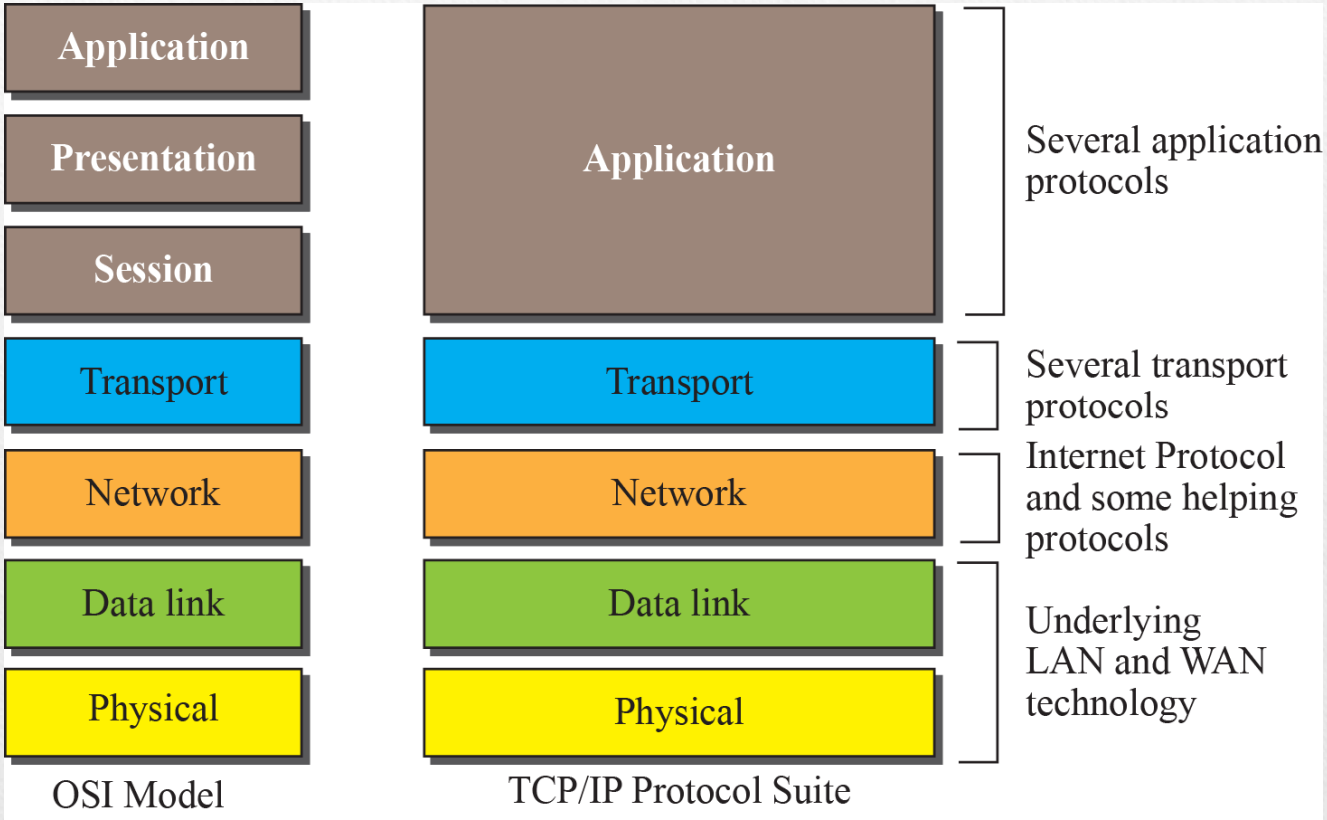


a. Multiplexing at source

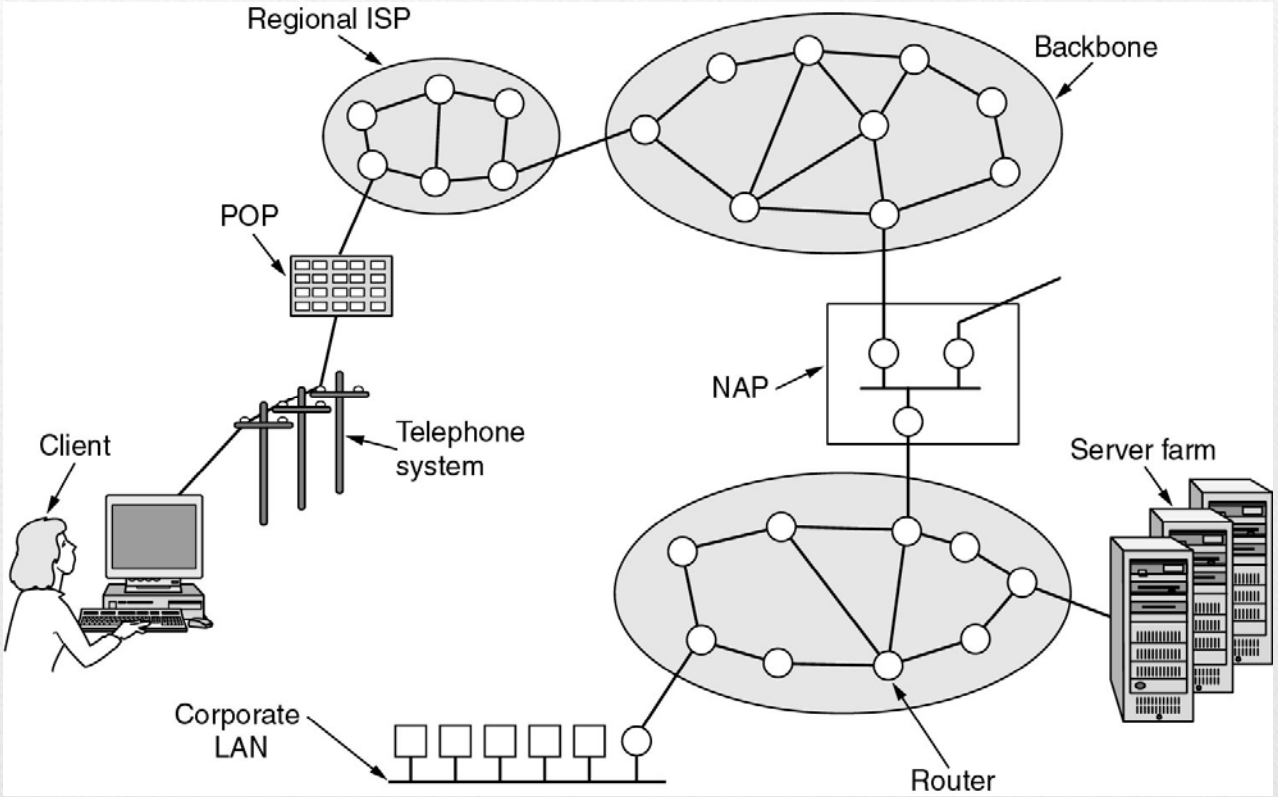


b. Demultiplexing at destination

TCP/IP and OSI Model



TCP/IP Reference Model



Next Class

- Physical Layer

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